Case Study: NimbusCart Checkout Redesign

Overview

This case study examines the 2025 checkout redesign for NimbusCart, a fictional mid-size e-commerce platform used for internal training purposes. The redesign project ran from March 2025 to September 2025 and involved a cross-functional team of 14 people, including designers, engineers, and data analysts.

Problem Statement

Prior to the redesign, NimbusCart's checkout flow had an average cart abandonment rate of 68.4%, notably higher than the industry benchmark of 55%. User research identified three primary friction points: a mandatory account-creation step, a five-page checkout process, and limited payment method options.

Methodology

The team ran a six-week discovery phase using surveys and session recordings from 2,300 users. This was followed by three rounds of A/B testing on a redesigned two-page checkout flow that introduced guest checkout and added support for four new payment methods, including a regional wallet called PayLoop.

Results

After the redesign launched in October 2025, cart abandonment dropped from 68.4% to 41.2% within eight weeks. Average checkout completion time fell from 3 minutes 40 seconds to 1 minute 12 seconds. Mobile conversion specifically improved by 37%, compared to an 18% improvement on desktop.

Key Takeaways

The case study concludes that removing mandatory account creation was the single highest-impact change, accounting for an estimated 60% of the total abandonment reduction. The project team recommended that future redesigns prioritize guest checkout options before investing in additional payment method integrations.

Limitations

This case study is a fictional scenario created for demonstration and testing purposes. Any resemblance to real companies, products, or datasets is coincidental. All figures are illustrative and were generated for the purpose of testing document-based research tools.